

Dhruvesh Chauhan

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EDUCATION

Rutgers University

Master's in Computer and Information Sciences

01/2025 - 12/2026

Mumbai University

Bachelor of Engineering in Information Technology

08/2017 - 05/2022

PROJECTS

Agentic Investment Analytics Platform (MCP & RAG)

06/2026 – PRESENT

- **Architected an MCP-driven AI financial platform** integrating Python, FastAPI, and React, deploying 7+ MCP tools to offload mathematical modeling from LLMs to a robust backend.
- **Developed a production-grade RAG pipeline** for SEC 10-K analysis using **AWS RDS PostgreSQL/pgvector**, executing semantic chunking and reranking to generate highly cited, grounded financial insights.
- **Optimized data ingestion pipelines, reducing multi-ticker processing latency by 46% (195s to 105s)** through asyncio concurrency, parallel embedding batches, and resolving asynchronous event loop bottlenecks.
- **Engineered and deployed a multi-tenant session architecture on AWS EC2**, synchronizing out-of-process MCP and REST APIs to enable real-time, link-isolated dashboard polling without authentication friction.
- **Built a deterministic XBRL financial analysis engine** that automates the extraction and integrity validation of multi-year financial statements, bypassing LLM hallucination for accurate Discounted Cash Flow (DCF) modeling and comparative analysis.
- **Demo:** <https://finsight-mcp-app.onrender.com/>

Agentic Text-to-SQL Reporting Engine

03/2026

- **Architected an AI-driven text-to-SQL pipeline** utilizing Model Context Protocol (MCP) and Groq, empowering non-technical stakeholders to autonomously query relational databases.
- **Engineered a dynamic schema-injection pipeline** that feeds human-readable data dictionaries into the LLM context window, drastically reducing query hallucinations and SQL syntax errors.
- **Developed a Human-in-the-Loop (HITL) security architecture** within the Flask backend, enforcing administrative UI approval for generated queries to prevent SQL injection and destructive database operations.
- Automated the complete data extraction lifecycle from natural language prompt to finalized Excel download, eliminating the need for developer tickets and reducing report generation time from days to minutes.
- **Demo:** <https://suspicious-reports-ui.onrender.com/> | [Video](#)

WORK EXPERIENCE

Associate Product Analyst | TSS Consultancy, Mumbai

10/2023– 12/2024

- **Contributed to the technical development of high-scale transaction monitoring tools** for major fintech clients (Zerodha, Upstox, Groww) by defining core backend business logic alongside engineering teams.
- **Translated complex financial regulatory mandates (SEBI, NSE) into precise technical specifications**, coordinating with developers to ensure the compliant and timely delivery of critical platform features.

CGI | Rutgers MBS Externship | New Brunswick, NJ

09/2025– 12/2025

- **Directed a 6-member engineering team** to build an AI-powered SDLC automation tool, managing agile task allocation and serving as the primary technical liaison for project stakeholders.
- Architected a custom RAG pipeline integrating the Jira API, Ollama, and ChromaDB to translate ticket intents into precise, vector-based codebase retrievals.
- **Automated end-to-end developer workflow**, from issue extraction to staging GitHub commits, enforced by a Human-in-the-Loop (HITL) verification layer to ensure safe code modification.

SKILLS

Languages: Python, TypeScript/JavaScript, SQL, Java

AI & Data Tech: Model Context Protocol (MCP), RAG, Hugging Face, Cursor

Frameworks: FastAPI, React, Spring Boot, Flask, Tailwind CSS

Databases: PostgreSQL (pgvector), MongoDB

Cloud & DevOps: AWS (EC2, RDS), GCP, Docker, CI/CD Pipelines, Git